Deterministic Optimization

Linear Programming Duality

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Lagrangian Relaxation and LP Duality

LP Duality

Learning Objectives for this lesson

- Discover the process of Lagrangian relaxation
- Apply Lagrangian relaxation to form LP dual

Systematic Ways to Carry Out Relaxation

- Let's illustrate with an example

$$
(P) \qquad Z_P = \min \quad x_1 + 2x_2 + 3x_3
$$

$$
\begin{aligned}
\text{s.t.} \quad & x_1 + 5x_2 + 4x_3 \geq 6 \\
& 2x_1 + 3x_2 - x_3 = 3 \\
& x_1 + x_2 - 2x_3 \leq 4 \\
& x_1 \geq 0, x_2 \leq 0, x_3 \text{ free}
\end{aligned}
$$

- **Step 1: Relax the objective function**

$$
x_1 + 2x_2 + 3x_3 + \underbrace{y_1 \cdot (6 - x_1 - 5x_2 - 4x_3)}_{\boxed{\leq 0}} + \underbrace{y_2 \cdot (3 - 2x_1 - 3x_2 + x_3)}_{\boxed{\leq 0}} + \underbrace{y_3 \cdot (4 - x_1 - x_2 + 2x_3)}_{\boxed{\leq 0}}
$$

Step 1: Relax Objective Function

- Step 1 (cont.)

$$
\boxed{\begin{array}{rcl}
x_1 + 5x_2 + 4x_3 \geq 6 & \rightarrow & y_1 \geq 0 \ \text{(because we want } y_1 \cdot (6 - x_1 - 5x_2 - 4x_3) \leq 0) \\
2x_1 + 3x_2 - x_3 = 3 & \rightarrow & y_2 \text{ free } \text{(because we want } y_2 \cdot (3 - 2x_1 - 3x_2 + x_3) \leq 0) \\
x_1 + x_2 - 2x_3 \leq 4 & \rightarrow & y_3 \leq 0 \ \text{(because we want } y_3 \cdot (4 - x_1 - x_2 + 2x_3) \leq 0).
\end{array}}
$$

- A new problem:

$$
\begin{aligned}
(B) \qquad \hat{Z}(\boldsymbol{y}) = \min \quad & (x_1 + 2x_2 + 3x_3) + y_1 \cdot (6 - x_1 - 5x_2 - 4x_3) + y_2 \cdot (3 - 2x_1 - 3x_2 + x_3) \\
& + y_3 \cdot (4 - x_1 - x_2 + 2x_3) \\
\text{s.t.} \quad & x_1 + 5x_2 + 4x_3 \geq 6 \\
& 2x_1 + 3x_2 - x_3 = 3 \\
& x_1 + x_2 - 2x_3 \leq 4 \\
& x_1 \geq 0, x_2 \leq 0, x_3 \text{ free.}
\end{aligned}
$$

Step 2: Relax Feasible Region

- **Step 2: Relax the feasible region by ignoring constraints:**

$$
\begin{aligned}
(LR) \quad Z(\boldsymbol{y}) = \min_{x_1, x_2, x_3} \quad & (x_1 + 2x_2 + 3x_3) + y_1 \cdot (6 - x_1 - 5x_2 - 4x_3) + y_2 \cdot (3 - 2x_1 - 3x_2 + x_3) \\
& + y_3 \cdot (4 - x_1 - x_2 + 2x_3) \\
\text{s.t.} \quad & x_1 \geq 0, x_2 \leq 0, x_3 \text{ free.}
\end{aligned}
$$

- This problem is called *the Lagrangian Relaxation Problem*
- Important relation:

$$Z(\boldsymbol{y}) \leq \hat{Z}(\boldsymbol{y}) \leq Z_P^* \text{ for any } y_1 \geq 0, \ y_2 \text{ free, and } y_3 \leq 0$$

Step 2 (cont.)

- Solve the Lagrangian relaxation problem:
  - An important structure of the LR problem is **Separability**

$$Z(\boldsymbol{y}) = \min_{x_1 : x_1 \geq 0} \{(1 - (y_1 + 2y_2 + y_3))\, x_1\} + \min_{x_2 : x_2 \leq 0} \{(2 - (5y_1 + 3y_2 + y_3))x_2\}$$
$$+ \min_{x_3 : x_3 \text{free}} \{(3 - (4y_1 - y_2 - 2y_3))x_3\} + (6y_1 + 3y_2 + 4y_3).$$

  - Each subproblem can be solved by hand in closed form

$$\min_{x_1 : x_1 \geq 0} \{(1 - (y_1 + 2y_2 + y_3))x_1\} = \begin{cases} 0 & \text{if } 1 - (y_1 + 2y_2 + y_3) \geq 0 \\ -\infty & \text{otherwise} \end{cases} \quad \text{(a)}$$

$$\min_{x_2 : x_2 \leq 0} \{(2 - (5y_1 + 3y_2 + y_3))x_2\} = \begin{cases} 0 & \text{if } 2 - (5y_1 + 3y_2 + y_3) \leq 0 \\ -\infty & \text{otherwise} \end{cases} \quad \text{(b)}$$

$$\min_{x_3 : x_3 \text{ free}} \{(3 - (4y_1 - y_2 - 2y_3))x_3\} = \begin{cases} 0 & \text{if } 3 - (4y_1 - y_2 - 2y_3) = 0 \\ -\infty & \text{otherwise.} \end{cases} \quad \text{(c)}$$

Step 3

- Lagrangian relaxation problem can be solved as

$$Z(\boldsymbol{y}) = \begin{cases} 6y_1 + 3y_2 + 4y_3 & \text{if (a), (b), (c) hold} \\ -\infty & \text{otherwise.} \end{cases}$$

- **Step 3: Find the best lower bound**

$$\begin{aligned} Z_D = \max_{y_1, y_2, y_3} \quad & Z(\boldsymbol{y}) \\ \text{s.t.} \quad & y_1 \geq 0, y_2 \text{ free}, y_3 \leq 0, \end{aligned}$$

Dual LP

$$\begin{aligned} (D) \quad Z_D = \max_{y_1, y_2, y_3} \quad & 6y_1 + 3y_2 + 4y_3 \\ \text{s.t.} \quad & 1 - (y_1 + 2y_2 + y_3) \geq 0 \\ & 2 - (5y_1 + 3y_2 + y_3) \leq 0 \\ & 3 - (4y_1 - y_2 - 2y_3) = 0 \\ & y_1 \geq 0, y_2 \text{ free}, y_3 \leq 0. \end{aligned}$$

General Rules to Form LP Dual

Step 1: Relaxing the objective

1. Since $y_i(b_i - \boldsymbol{a}_i^\top \boldsymbol{x}) \leq 0$:
   - If $\boldsymbol{a}_i^\top \boldsymbol{x} \geq b_i$, then $y_i \geq 0$;
   - If $\boldsymbol{a}_i^\top \boldsymbol{x} \leq b_i$, then $y_i \leq 0$;
   - If $\boldsymbol{a}_i^\top \boldsymbol{x} = b_i$, then y_i free.

Step 2: Solving Lagrangian Relaxation

2. Since $(c_j - \boldsymbol{y}^\top \boldsymbol{A}_j)x_j \geq 0$:
   - If $x_j \geq 0$, then $c_j \geq \boldsymbol{y}^\top \boldsymbol{A}_j$ (i.e. $\boldsymbol{y}^\top \boldsymbol{A}_j \leq c_j$);
   - If $x_j \leq 0$, then $c_j \leq \boldsymbol{y}^\top \boldsymbol{A}_j$ (i.e. $\boldsymbol{y}^\top \boldsymbol{A}_j \geq c_j$);
   - If x_j free, then $c_j = \boldsymbol{y}^\top \boldsymbol{A}_j$ (i.e. $\boldsymbol{y}^\top \boldsymbol{A}_j = c_j$).

Primal and Dual Pair

$$
\begin{aligned}
(P)\quad \min\ & \sum_{j=1}^{n} c_j x_j \\
\text{s.t.}\ & \sum_{j=1}^{n} a_{ij} x_j \geq b_i \quad \text{for all } i = 1, \ldots, m_1 \\
& \sum_{j=1}^{n} a_{ij} x_j = b_i \quad \text{for all } i = m_1 + 1, \ldots, m_2 \\
& \sum_{j=1}^{n} a_{ij} x_j \leq b_i \quad \text{for all } i = m_2 + 1, \ldots, m \\
& x_j \geq 0 \quad \text{for all } j = 1, \ldots, n_1 \\
& x_j \text{ is free} \quad \text{for all } j = n_1 + 1, \ldots, n_2 \\
& x_j \leq 0 \quad \text{for all } j = n_2 + 1, \ldots, n.
\end{aligned}
$$

$$
\begin{aligned}
(D)\quad \max\ & \sum_{i=1}^{m} b_i y_i \\
\text{s.t.}\ & \sum_{i=1}^{m} a_{ij} y_i \leq c_j \quad \text{for all } j = 1, \ldots, n_1 \\
& \sum_{i=1}^{m} a_{ij} y_i = c_j \quad \text{for all } j = n_1 + 1, \ldots, n_2 \\
& \sum_{i=1}^{m} a_{ij} y_i \geq c_j \quad \text{for all } i = n_2 + 1, \ldots, n \\
& y_i \geq 0 \quad \text{for all } i = 1, \ldots, m_1 \\
& y_i \text{ is free} \quad \text{for all } i = m_1 + 1, \ldots, m_2 \\
& y_i \leq 0 \quad \text{for all } i = m_2 + 1, \ldots, m.
\end{aligned}
$$

Summary

- We learned:
  - Lagrangian relaxation procedure
  - Detailed steps carrying out Lagrangian relaxation for an example
  - General rules for forming LP dual